

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ | A description of all covariates tested |
| ☐ | ☒ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

- Flow cytometry data were collected using the BD FACSDiva software (version 9.0) or the SpectroFlo software (version 3.1.0, Cytex)
- ELISA data were collected using Clariostar Plus (BMG, Germany).
- DNA was quantified using DeNovix Nanodrop (Thermofisher).

Data analysis

- Flow cytometry data analysis was performed using FlowJo V10.5 and V10.8.
- Heatmap representations of flow cytometry data (percentages of positive cells for defined markers) were created with the MeV (Multiple Experiment Viewer) software.
- Percentages of editing after CRISPR/Cas9 knockout were assessed using the ICE CRISPR Analysis Tool V2 (Synthego).
- LEGENDplex™ data were analyzed using the LEGENDplex™ Data Analysis Software (<https://www.biolegend.com/de-at/immunoassays/legendplex>).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All key datasets generated and analyzed in this study are available in the Source Data file provided. Additional data supporting the findings of this study are available from the corresponding authors upon reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Findings apply to individuals of all sexes and genders. Sex and gender were not considered in the design of the SPARCKLING and HORUS studies, and were self-reported. As CMV infection can affect all sexes and genders and sex is not a known risk factor, we did not perform sex- or gender- based analyses.
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	Included patients were all kidney transplant recipients with a clinically significant CMV infection requiring treatment. They were included either in the SPARCKLING (ClinicalTrials.gov Identifier: NCT03339661) or the HORUS study (NCT05701228). More information regarding these two studies are available at ClinicalTrials.gov
Recruitment	Patients were included by a transplant physician with the help of a study coordinator. Briefly, they were included at the onset of a CMV infection requiring treatment. For this specific study we included only patients with recurrent or refractory CMV infections based on the most recent definition.
Ethics oversight	Kidney transplant recipients were included in the SPARCKLING study (NCT03339661), which was approved by an ethics committee (Bordeaux University Hospital, CHUBX 2016/40), or the HORUS study (NCT05701228), which was also approved by an ethics committee (Bordeaux University Hospital, CHUBX 2022/10). Blood samples from healthy donors and kidney transplant recipients were obtained after written informed consent, in accordance with the Declaration of Helsinki

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for experiments involving healthy human donors were not determined using statistical power calculations but were instead based on prior experience and experimental observations to ensure data robustness and reproducibility. For in vivo experiments, sample sizes were determined upfront based on prior experimental data to detect meaningful differences while complying with ethical guidelines. For experiments involving patient samples, sample size was determined by the number of eligible participants that could be recruited during the study period. Sample sizes for each experiment are provided in the corresponding figure legends.
Data exclusions	No data were excluded from the analysis.
Replication	To verify the reproducibility of the experimental findings: - to address experimental variations, all in vitro experiments were performed in duplicates or triplicates of culture. This information is indicated in the corresponding figure legends. - to address biological variations, all in vitro experiments were repeated on independent healthy donors or patients. This information is indicated in the corresponding figure legends. Each analysis presented in this study is based on a minimum of 3 independent experiments performed on independent donors or patients. These experiments were performed during different moments in time. All attempts at replications were successful.
Randomization	Mice used for in vivo experiments were randomly assigned to the different groups.

Blinding was not applicable to in vitro experiments. Well-controlled quantitative measurement methods were used to exclude bias. For in vivo experiments, the follow-up of MCMV-infected mice after gamma-delta T cell adoptive transfer was performed blinded to the conditions of the experiment.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

The following antibodies were used in this study (marker; clone; fluorochrome; catalog number; supplier; dilution):

Anti-human antibodies:
CD3, UCHT1, V450, 560365, BD Biosciences, 1:50
CD27, M-T271, APC, 558664, BD Biosciences, 1:16.5
CD45RA, REA562, VioGreen, 130-113-931, Miltenyi Biotec, 1:50
Vδ1 TCR, REA173, FITC, 130-100-532, Miltenyi Biotec, 1:50
Vδ2 TCR, REA771, PE-Vio770, 130-111-612, Miltenyi Biotec, 1:100
γδTCR, REA591, PE, 130-113-512, Miltenyi Biotec, 1:50
αβTCR, REA652, APC, 130-113-535, Miltenyi Biotec, 1:50
CD56, 5.1H11, BV421, 362551, Biolegend, 1:50
CD3, UCHT1, BV510, 563109, BD, Biosciences, 1:25
CD14, 63D3, BV711, 367139, Biolegend, 1:50
CD16, 3G8, BV786, 563690, BD Biosciences, 1:25
CD19, HIB19, PE-Cy5, 555414, BD Biosciences, 1:10
γδTCR, REA591, APC, 130-113-508, Miltenyi Biotec, 1:12.5
Vδ3, TCR, Uncoupled, 1563, Immunotech, 1:40
CD3, UCHT1, BV510, 563109, BD Biosciences, 1:25
Vδ2 TCR, REA771, VioGreen, 130-111-133, Miltenyi Biotec, 1:25
γδTCR, REA591, APC, 130-113-508, Miltenyi Biotec, 1:12.5
CD25, M-A251, BV421, 562443, BD Biosciences, 1:25
HLA-DR, G46-6, BV711, 563696, BD Biosciences, 1:25
CD71, M-A712, BV786, 563768, BD Biosciences, 1:25
CD69, FN50, AF700, 560739, BD Biosciences, 1:50
CD150, A12, BV421, 562875, BD Biosciences, 1:10
CD137, 4B4-1, BV650, 564092, BD Biosciences, 1:100
CD134, ACT35, BV711, 563664, BD Biosciences, 1:50
CD30, BERH8, BV786, 744411, BD Biosciences, 1:5
CD154, TRAP1, APC, 560955, BD Biosciences, 1:10
CD28, CD28.2, APC-R700, 565181, BD Biosciences, 1:25
CD226, DX11, BV421, 742493, BD Biosciences, 1:100
NKp46, 9E2/NKp46, BV650, 563230, BD Biosciences, 1:10
NKp30, p30-15, BV711, 563383, BD Biosciences, 1:10
NKG2D, 1D11, BV786, 743560, BD Biosciences, 1:25
NKp44, P44-8.1, AF647, 558564, BD Biosciences, 1:2.5
CD16, 3G8, AF700, 557920, BD Biosciences, 1:50
TIM-3, 7D3, BV711, 565566, BD Biosciences, 1:16.5
PD-1, EH12.1, BV650, 564104, BD Biosciences, 1:25
TIGIT, 741182, BV421, 747844, BD Biosciences, 1:50
CTLA-4, BNI3, APC, 560938, BD Biosciences, 1:5
LAG-3, T47-530, APC-R700, 565775, BD Biosciences, 1:10
TRAIL, RIK-2, BV421, 564243, BD Biosciences, 1:10

Perforin, δ G9, BV421, 563393, BD Biosciences, 1:10
 FasL, NOK-1, BV711, 744101, BD Biosciences, 1:10
 Granulysin, RB1, AF488, 558254, BD Biosciences, 1:2.5
 Granzyme, B, GB11, AF700, 560213, BD Biosciences, 1:10
 CD11a, (LFA-1), HI111, BV711, 565388, BD Biosciences, 1:16.5
 CD8 α , RPA-T8, APC, 555369, BD Biosciences, 1:2.5
 KIR3DL1 (CD158e), REA1005, VioBlue, 130-116-943, Miltenyi Biotec, 1:25
 KIR2DL1/S1 (CD158a/h), REA1010, VioBright, 515, 130-116-982, Miltenyi Biotec, 1:25
 KIR2DL2/L3 (CD158b), REA1006, PE, 130-116-952, Miltenyi Biotec, 1:50
 KIR2DS4 (CD158i), REA860, PE-Vio615, 130-114-780, Miltenyi Biotec, 1:25
 KIR2DL1 (CD158a), REA284, PE-Vio770, 130-120-587, Miltenyi Biotec, 1:50
 KIR3DL1/S1 (CD158e1/e2), REA168, APC-Vio770, 130-104-884, Miltenyi Biotec, 1:5
 NKG2C, REA205, APC, 130-117-547, Miltenyi Biotec, 1:25
 CD107a, H4A3, BV421, 566261, BD Biosciences, 1:40
 Human Cytomegalovirus IE1, 8B1.2, AF488, MAB810X, Merck Millipore, 1:50
 CD14, REA599, Vioblue, 130-094-364, Miltenyi Biotec, 1:10
 ICAM1 (CD54), HA58, BV421, 564077, BD Biosciences, 1:10
 Mouse IgG1, κ Isotype ctrl, X40, BV421, 562438, BD biosciences, 1:80

Anti-mouse antibodies:

CD3, 145-2C11, PE-CF594, 562332, BD Biosciences, 1:50
 $\gamma\delta$ TCR, GL3, BV421, 562892, BD Biosciences, 1:100
 CD11a (LFA-1), M17/4, BUV395, 740228, BD Biosciences, 1:200
 CD25, PC61, BB515, 564458, BD Biosciences, 1:25
 CD44, IM7, V500, 560780, BD Biosciences, 1:100
 CD62L, MEL-14, PerCp-Cy5.5, 560513, BD Biosciences, 1:50
 CD178 (FasL), REA1171, PE-Vio770, 130-120-930, Miltenyi Biotec, 1:25
 CD226 (DNAM-1), 10E5, APC, 128809, Biolegend, 1:50
 CD253 (TRAIL), REA1030, PE, 130-117-560, Miltenyi, Biotec 1:25
 CD279 (PD-1), J43, BV605, 563059, BD Biosciences, 1:25
 CD314 (NKG2D), CX5, BV711, 563694, BD Biosciences, 1:50
 CD366 (TIM-3), 25F.1D6, RB780, 569884, BD Biosciences, 1:25

Anti-human antibodies used in blocking and activation experiments (target; clone; catalog number; supplier; working concentration):

Mouse IgG1, κ isotype control, MOPC-21, 400124, Biolegend, 10 μ g/ml
 TCR V δ 1, TCS-1, #TCR1055, Invitrogen, 10 μ g/ml
 NKp30, P30-15, 325202, Biolegend, 10 μ g/ml
 NKp44, P44-8, 325102, Biolegend, 10 μ g/ml
 NKG2D, 1D11, 320810, Biolegend, 10 μ g/ml
 DNAM-1, DX11, 559787, BD Biosciences, 10 μ g/ml
 CD54 (ICAM-1), HCD54, 322703, Biolegend, 10 μ g/ml
 CD11a (LFA-1) for blocking, HI111, 301234, Biolegend, 10 μ g/ml
 CD11a (LFA-1) for activation, 25.3, IM0157, Beckman/ImmuneTech, 0.1 μ g/ml to 10 μ g/ml

Validation

All flow cytometry antibodies used in this study were validated by the manufacturers, as detailed on the datasheets available after referencing antibody catalog numbers on their homepage:

- BD Biosciences: <https://www.bdbiosciences.com/en-fr/products/reagents/flow-cytometry-reagents/research-reagents>
- Biolegend: <https://www.biolegend.com/de-at/antibodies-reagents/primaries>
- Miltenyi Biotec: <https://www.miltenyibiotec.com>
- Invitrogen: <https://www.thermofisher.com/fr/fr/home.html>
- Beckman/ImmuneTech: <https://www.beckman.fr>
- Merck Millipore: <https://www.merckmillipore.com>

Anti-human flow cytometry antibodies were titrated to obtain an optimal staining index using PBMCs or DOT cells obtained from healthy donors. Anti-mouse flow cytometry antibodies were titrated using splenocytes or in vitro expanded mouse $\gamma\delta$ T cells obtained from C57BL/6J wild-type mice.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

- Human foreskin fibroblasts (primary cells) were provided by Dr. Hamid-Reza Rezvani (INSERM, U1035, Bordeaux).
- Conditionally immortalized glomerular endothelial cells (CiGEnC, <https://doi.org/10.1038/sj.ki.5000277>) were provided by Pr Dany Anglicheau (Department of Nephrology and Kidney Transplantation, Necker Hospital, Assistance Publique-Hôpitaux de Paris, Paris, France).
- MRC5 cells were obtained from Eurobio Scientific (ATCC CCL-171).

Authentication

MRC5 cell line authentication was provided by the supplier.

Mycoplasma contamination

All cell lines were regularly tested negative for mycoplasma contamination.

Commonly misidentified lines
(See [ICLAC](#) register)

No commonly misidentified cell lines were used in this study.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

C57BL/6J mice were purchased from Charles River Laboratories and Rag-/- γ c-/- mice were obtained from the Centre de Distribution, Typage et Archivage animal (CDTA), Orleans, France. Mice were bred and maintained under specific pathogen-free conditions in an appropriate biosafety level 2 facility (Bordeaux University, approval n° B33-063-916). Mice were housed under controlled conditions (21°C, 55-65% humidity, 12h/12h light/dark cycle). Experiments were performed on 8-12 weeks' old mice, in accordance with governmental guidelines and european ethical regulations. The in vivo protocols described in this study were notably approved by a local ethics committee (Bordeaux University committee n°2016092917471799).

Wild animals

No wild animals were used in this study

Reporting on sex

Sex was not considered as a factor in any experiment or analysis reported in this study.

Field-collected samples

No field-collected samples were involved in this study.

Ethics oversight

In vivo experiment were performed in accordance with governmental guidelines and european ethical regulations. The in vivo protocols described in this study were all approved by a local ethics committee (n°2016092917471799).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

Patients were included in the SPARCKLING study (ClinicalTrials.gov Identifier: NCT03339661), which was approved by an ethics committee (Bordeaux University Hospital, CHUBX 2016/40), or the HORUS study (NCT05701228), also approved by an ethics committee (Bordeaux University Hospital, CHUBX 2022/10).

Study protocol

Please see information provided at ClinicalTrials.gov (NCT03339661 and NCT05701228).

Data collection

SPARCKLING (from 2017-11-23 to 2020-11-23), HORUS (2023-06-26 to 2025-06)

Outcomes

The predefined primary outcome in this study was the ability to expand DOT cells from kidney transplant recipients. The secondary outcome was the ability of DOT cells to respond to CMV infection in vitro.

Plants

Seed stocks

N/A

Novel plant genotypes

N/A

Authentication

N/A

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

For phenotyping of PBMCs, sorted monocytes or DOT cells (obtained from healthy donor or kidney transplant recipients' peripheral blood): single cell suspensions were washed twice in staining buffer (PBS supplemented with 0.5% BSA and 2mM EDTA), stained with a viability dye (Fixable Viability Stain 780, BD Biosciences or Viability 405/520 Fixable Dye, Miltenyi Biotec), washed twice in staining buffer, stained with antibodies directed against surface markers (described in the Methods section and supplementary tables) for 20 minutes at 4°C in the dark, washed twice in staining buffer and analyzed. The same sample preparation protocol was used for DOT cell sorting after gamma-delta TCR knockout.

For degranulation assays, DOT cells were incubated 4 hours before harvesting and staining with an anti-CD107a antibody, 0.67 µL/mL monensin (BD GolgiStop, BD Biosciences) and 1 µg/mL brefeldin A (Sigma).

For dissemination assays, fibroblasts were washed in PBS, detached using Accutase Cell Detachment Solution (Sigma), washed in staining buffer and analyzed for GFP expression.

For the analysis of CMV infection in fibroblasts, macrophages or endothelial cells, cells were washed in PBS, detached using Accutase Cell Detachment Solution (Sigma), washed, fixed and permeabilized using the Foxp3/Transcription Factor Staining Buffer Set (eBioscience) according to the manufacturer's instructions, stained with an antibody directed against the human CMV Immediate-Early antigen (Merck Millipore) for 30min at room temperature in the dark, washed and analyzed.

For mouse gamma-delta T cell isolation and characterization, splenocytes were isolated through 70 µm cell strainers, followed by red blood cell lysis in ACK lysis buffer, washed and subsequently magnetically-depleted of αβ T cells and NK cells. Single cell suspensions of splenocytes and expanded mouse gamma-delta T cells were washed twice in staining buffer, stained with a viability dye (Fixable Viability Stain 780, BD Biosciences), washed twice in staining buffer, stained with antibodies directed against surface markers (described in the Methods section and supplementary tables) for 20 minutes at 4°C in the dark, washed twice in staining buffer and analyzed.

For the analysis of gamma-delta T cell infiltration in spleens, lungs and livers: spleens were processed as described above while lungs were digested at 37°C during 40min in RPMI 1640 supplemented with 1 mg/mL collagenase I and 50 µg/mL DNase I (Sigma), then passed through 70 µm cell strainers. Livers were directly dissociated on 70 µm cell strainers. After red blood cell lysis, lung and liver cell suspensions were separated by a discontinuous 40/80% Percoll gradient (according to manufacturer's instructions, Sigma). Immune cells were then harvested and stained as described above.

Instrument

For cell analysis: Canto II cytometer (BD Biosciences); LSR II Fortessa cytometer (BD Biosciences); Aurora cytometer (Cytek)
For cell sorting: FACS ARIA (BD Biosciences).

Software

For the cytometers from BD Biosciences, flow cytometry data was collected using the BD FACSDIVA software (version 9.0).
For the Aurora cytometer from Cytek, flow cytometry data was collected using the SpectroFlo® software (version 3.1.0).
FlowJo v10.5 or v10.8. was used for flow cytometry data analysis.

Cell population abundance

gdTCR knockout DOT cells were stained with an anti-gdTCR antibody and sorted to eliminate remaining gdTCR-positive cells. After sorting, >99% of DOT cells were negative for gdTCR expression (based on gating using control gdTCR-positive cells).

Gating strategy

For PBMC and DOT cell analysis in the context of DOT cell expansion, gating strategies are provided in Figure 1A and Supplementary Figure 1. Cells were gated as follows:

- 1) FSC-A vs SSC-A to gate on lymphocytes.
- 2) FSC-A vs FSC-W or FSC-H vs FSC-W to gate on singlets.
- 3) Viability dye vs SSC-A to gate on live cells (viability dye-negative cells).
- 4) gdTCR vs CD3 to gate on gd T cells (gdTCR+ CD3+).
- 5) Vd1 TCR vs Vd2 TCR to analyze the proportions of Vd1+; Vd2+ or Vd1negVd2neg gd T cells. When indicated, total Vd2neg gd T cells were gated for further analysis of surface markers.

The % of Vd2neg gd T cells positive for surface markers were determined using Fluorescence Minus One controls.

For KIR expression analysis, gating strategies are provided in Supplementary Figure 2. Cells were gated as follows:

- 1) FSC-A vs SSC-A to gate on lymphocytes.
- 2) FSC-H vs FSC-W followed by SSC-H vs SSC-W to gate on singlets.
- 3) SSC-A vs Viability dye and Vd2 TCR to exclude both dead cells and Vd2+ gd T cells.
- 4) SSC-A vs gdTCR to create two gates: one for gdTCRneg cells and one for gdTCR+ cells (Vd2neg gd T cells). KIR expression was then analyzed on both gdTCRneg and gdTCR+ cells.

For the analysis of CMV-infection in fibroblasts, macrophages and endothelial cells:

- 1) FSC-A vs SSC-A to exclude debris.
- 2) FSC-H vs FSC-W to gate on singlets.
- 3) SSC-A vs CMV-IE to gate on CMV-IE negative (non-infected) or CMV-IE positive (infected) cells. Mock-infected cultures were analyzed in parallel to set the gate for CMV-IE positivity.

For the analysis of DOT cell degranulation:

- 1) FSC-A vs SSC-A to exclude debris.

- 2) FSC-H vs FSC-W followed by SSC-H vs SSC-W to gate on singlets.
 - 3) Viability dye vs SSC-A to gate on live cells (viability dye-negative cells).
 - 4) CFSE vs SSC-A to gate on CFSE-labeled DOT cells.
 - 5) gdTCR vs Vd2 TCR to gate on gdTCR+ Vd2 TCRneg gd T cells.
- The percentages of Vd2neg gd T cells positive for CD107a were determined using Fluorescence Minus One controls.

For the analysis of CMV dissemination, the gating strategy is provided in Supplementary Figure 4E:

- 1) FSC-A vs SSC-A to gate on fibroblasts and exclude debris and DOT cells.
- 2) GFP vs SSC-A to gate on GFP+ (CMV-infected) fibroblasts. Mock-infected cultures were analyzed in parallel to set the gate for GFP positivity.

For the analysis of splenocytes and in vitro expanded mouse gamma-delta T cells, gating strategy is provided in Figure 4A:

- 1) FSC-A vs SSC-A to gate on lymphocytes.
- 2) FSC-A vs FSC-H to gate on singlets.
- 3) FSC-A vs viability dye to gate on live cells (viability dye-negative).
- 4) CD3 vs gdTCR to gate on gd T cells (CD3+ gdTCR+).

The percentages of gd T cells positive for surface markers were further analyzed. CD62L vs CD44 gating was used to analyze naive (CD62L+ CD44neg), TCM (CD62L+ CD44+) and TEM (CD62Lneg CD44+) gd T cells.

For the analysis of mouse gamma-delta T cell infiltration in organs after adoptive transfer:

- 1) FSC-A vs SSC-A to gate on lymphocytes.
- 2) FSC-A vs FSC-H to gate on singlets.
- 3) FSC-A vs viability dye to gate on live cells (viability dye-negative).
- 4) CD3 vs gdTCR to gate on gd T cells (CD3+ gdTCR+).
- 5) CD62L vs CD44 to gate on naive (CD62L+ CD44neg), TCM (CD62L+ CD44+) and TEM (CD62Lneg CD44+) gd T cells.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.